Lecture 1: Inverse Functions and Their Derivatives

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Determine if an inverse function exists
- ☐ Find the inverse of a function.
- ☐ Find the derivative of the inverse function.

Preliminary Examples

Example 1: Let $f(x) = 2x + 1$ with the domain $(-\infty, \infty)$, we wish to find a function that "undoes" f. Or in another words, we set $y = f(x) = 2x + 1$, we want to solve for x in terms of y.

Let $y = 2x + 1$

$\Rightarrow 2x = y - 1$

$\Rightarrow x = \frac{y-1}{2} = g(y)$

This function g "undoes" the function f where

$g(y) = \frac{y-1}{2} = x$

Preliminary Examples

Example 2: Let $f(x) = x^2$ with the domain $(-\infty, \infty)$, and we wish to find a function that "undoes" f. Similarly, we set $y = f(x) = x^2$ and we try to solve for x in terms of y.

Let $y = x^2$

$\Rightarrow x = \pm\sqrt{y}$ or $x = \sqrt{y}$ or $-\sqrt{y}$

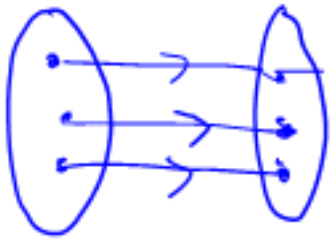
We cannot find one particular function that will "undo" the function f.

Why? For any particular value of y except 0, there are two possible values that $f(x)$ could have come from.

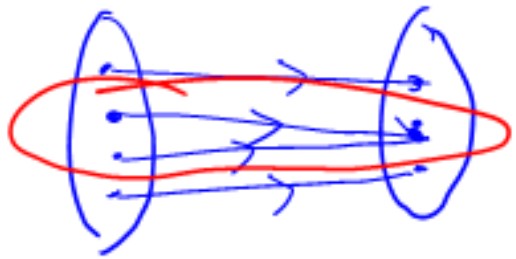


One-to-one Functions

A function is said to be one-to-one (injective) if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ (or in other words, $f(x_1) = f(x_2)$ implies $x_1 = x_2$) for all x_1, x_2 in the domain of f.



1-1 function



not a 1-1 function.

Example: $f(x) = x + 1$ is one-to-one, but $g(x) = x^2 + 1$ is not one-to-one.

Let $f(x_1) = f(x_2)$ for $x_1, x_2 \in$ Domain.

$\Rightarrow x_1 + 1 = x_2 + 1$

$\Rightarrow x_1 = x_2$

So f is 1-1

$g(1) = 1^2 + 1 = 2$

$g(-1) = (-1)^2 + 1 = 2$

$g(1) = g(-1)$ but $1 \neq -1$

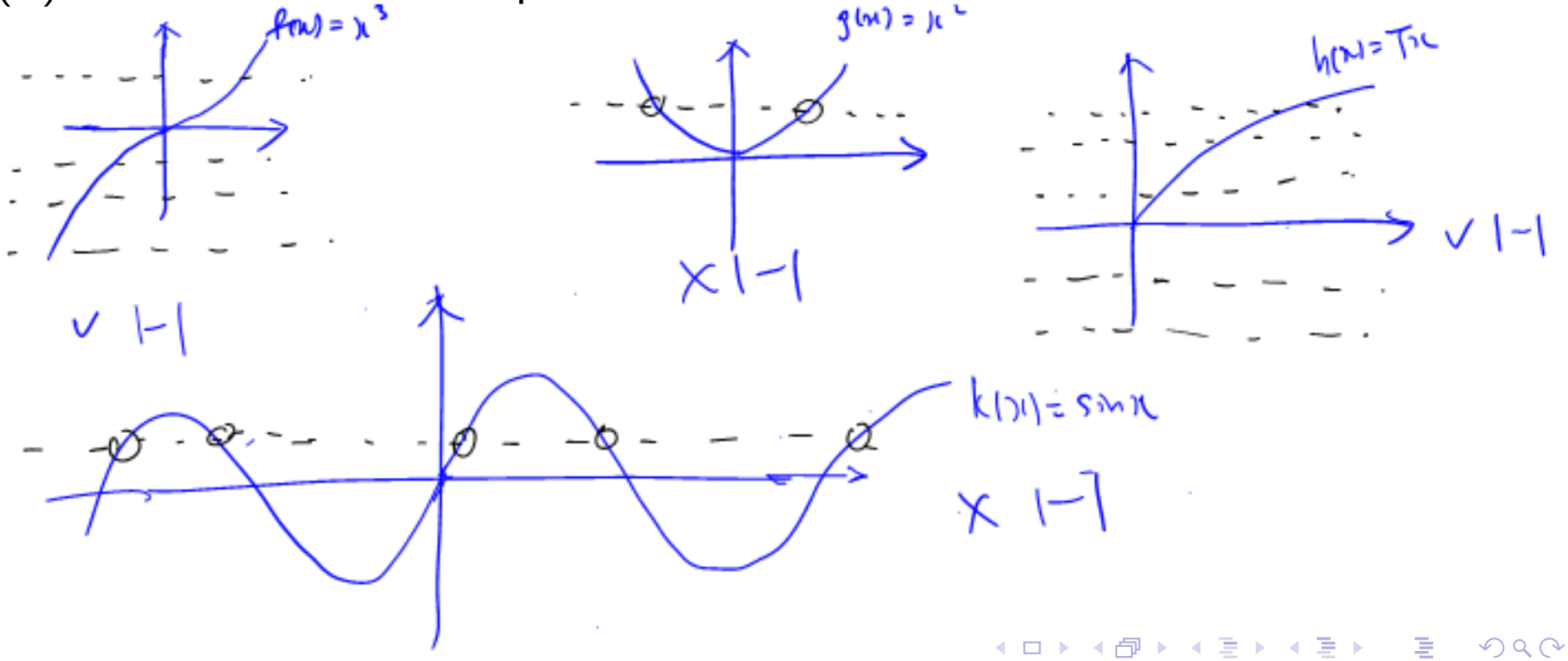
So g is not 1-1.

One-to-one Functions

Horizontal Line Test

A function is said to be one-to-one if and only if its graph intersect each horizontal line at most once. 0/1

Example: Consider the graphs $f(x) = x^3$, $g(x) = x^2$, $h(x) = \sqrt{x}$ and $k(x) = \sin x$. Which ones pass the horizontal line test?

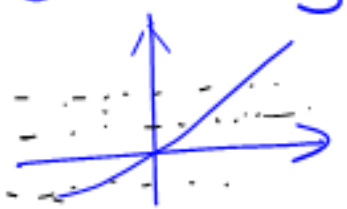


One-to-one Functions

If a function f is either strictly increasing or strictly decreasing on its domain, then f is one-to-one.

strictly increasing $\Rightarrow$ If $x_2 > x_1$, then $f(x_2) \geq f(x_1)$, $f'(x) > 0 \ \forall x$

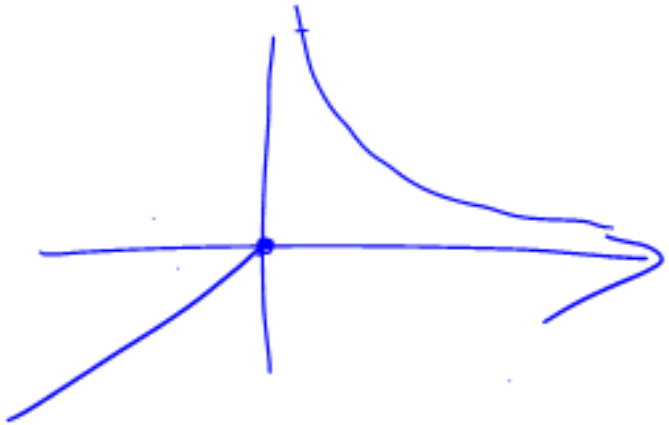
strictly decreasing $\Rightarrow$ If $x_2 > x_1$, then $f(x_2) < f(x_1)$, $f'(x) < 0 \ \forall x$.



1-1

Be aware, the converse is not true, there is one-to-one function that is neither strictly increasing nor strictly decreasing.

Eg:



$$f(x) = \begin{cases} x, & x \leq 0 \\ \frac{1}{x}, & x > 0 \end{cases}$$

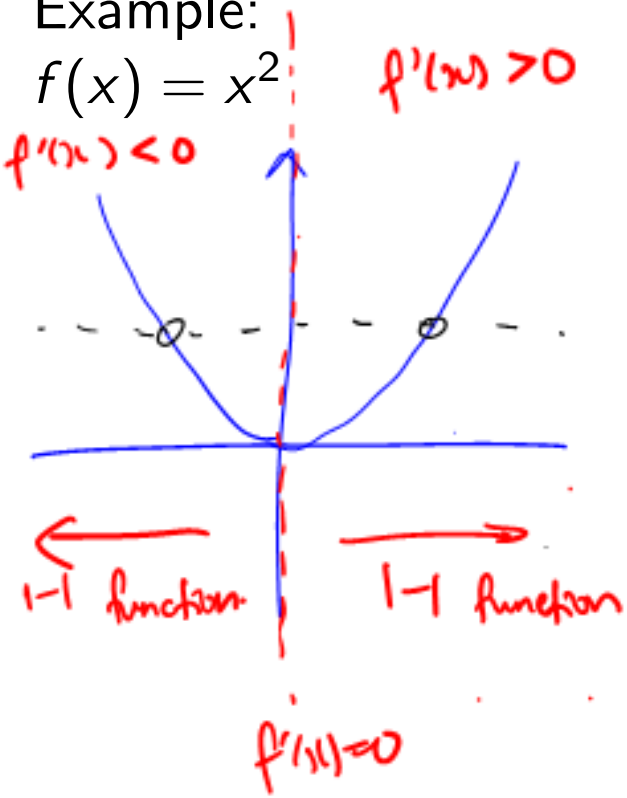
1-1 but the graph is neither strictly increasing nor strictly decreasing.

One-to-one Functions

Not all functions are one-to-one on their entire natural domain. However, by restricting the function to a smaller domain, we can create a function that is one-to-one.

Example:

$f(x) = x^2$



$I_1 = (-\infty, 0]$

$I_2 = [0, \infty)$

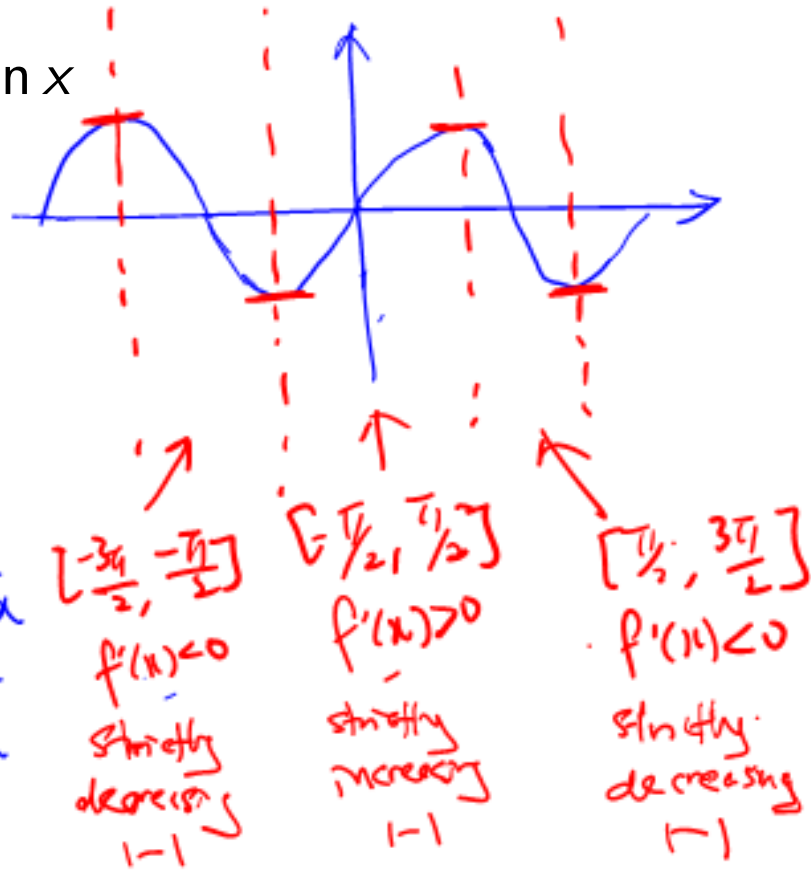
f is 1-1 on I_1

f is 1-1 on I_2

on I_1, $f^{-1}(x) = -\sqrt{x}$

on I_2, $f^{-1}(x) = \sqrt{x}$

$g(x) = \sin x$



Inverse Functions

Suppose that f is a one-to-one function on a domain D with range R. Then there exists a unique function g satisfying

$$g(f(x)) = x \text{ for all } x \text{ in the domain of } f, \text{ and}$$
$$f(g(x)) = x \text{ for all } x \text{ in the range of } f.$$

Furthermore, $\text{Domain}(g) = \text{Range}(f)$ and $\text{Range}(g) = \text{Domain}(f)$ and g is also one-to-one.

This unique function g is called the inverse function of f and is denoted by f^{-1}.

Only one-to-one function can have an inverse.

If f is strictly increasing or strictly decreasing, then f is one-to-one and has an inverse.

Finding Inverses

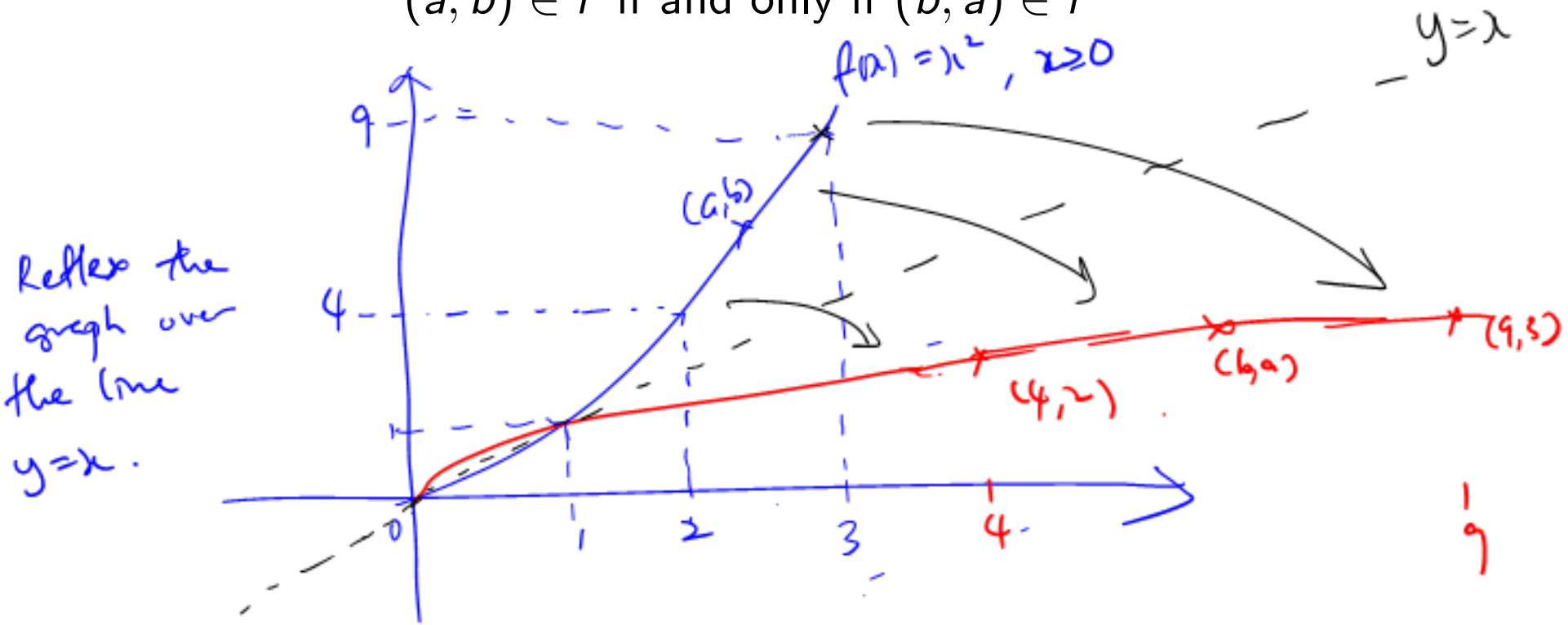
$f^{-1}(y) = x$

~~$f^{-1}(x) = y$~~ if and only if $y = f(x)$

In another words,

$(2,4) \in f$ $(4,2) \in f^{-1}$

$(a, b) \in f$ if and only if $(b, a) \in f^{-1}$



Finding Inverses

Example: Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x.

Let $f(x) = \frac{1}{2}x + 1$

f is strictly increasing $\Rightarrow f$ is 1-1 and has an inverse.

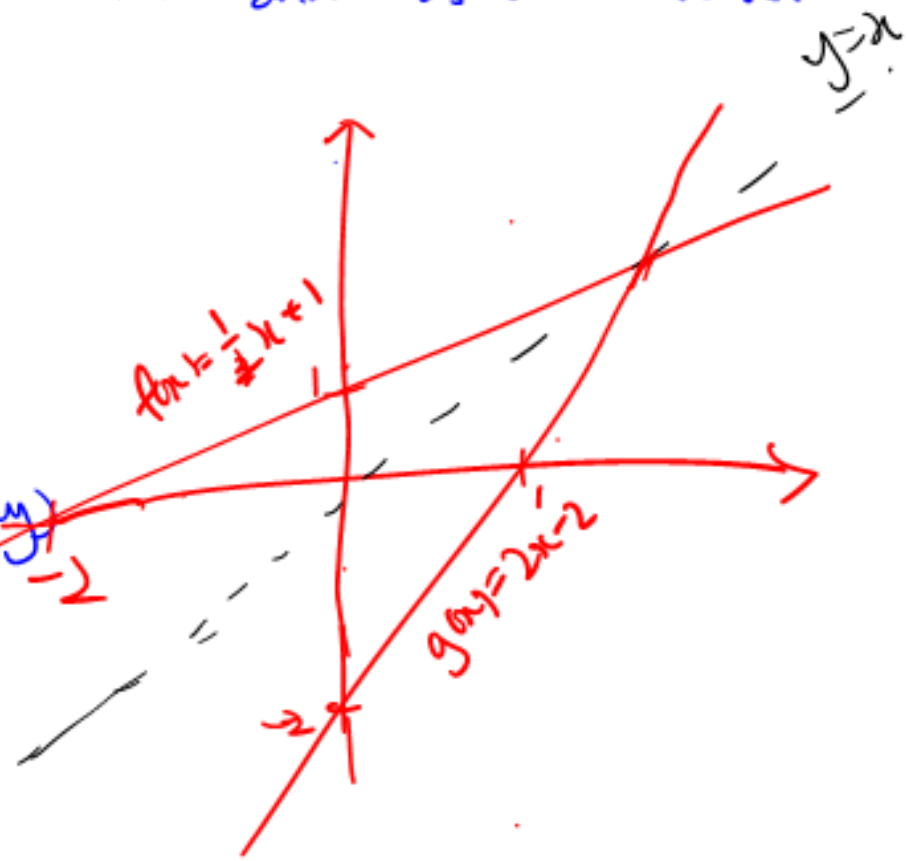
Let $y = \frac{1}{2}x + 1$

Solve x in terms of y.

$\frac{1}{2}x = y - 1$

$x = 2(y-1) = 2y - 2 = g(y)$

So, $f^{-1}(x) = g(x) = 2x - 2$.



Finding Inverses

Example: Find the inverse of $y = x^2$, $x \geq 0$, expressed as a function of x.

At $x \geq 0$, $f(x) = x^2$ is strictly increasing.

So f is 1-1 and has an inverse.

Let $y = x^2$

Solve x in terms of y.

$x = -\sqrt{y}$ or $\sqrt{y}$ ($x \geq 0$)

(× for $-\sqrt{y}$, ✓ for $\sqrt{y}$)

$\Rightarrow x = \sqrt{y} = g(y)$

So $f^{-1}(x) = g(x) = \sqrt{x}$

Derivatives of Inverses of Differentiable Functions

Suppose that the ~~derivative~~ inverse of ~~an~~ a ~~invertible~~ differentiable function exists, under which circumstances is the inverse function also differentiable?

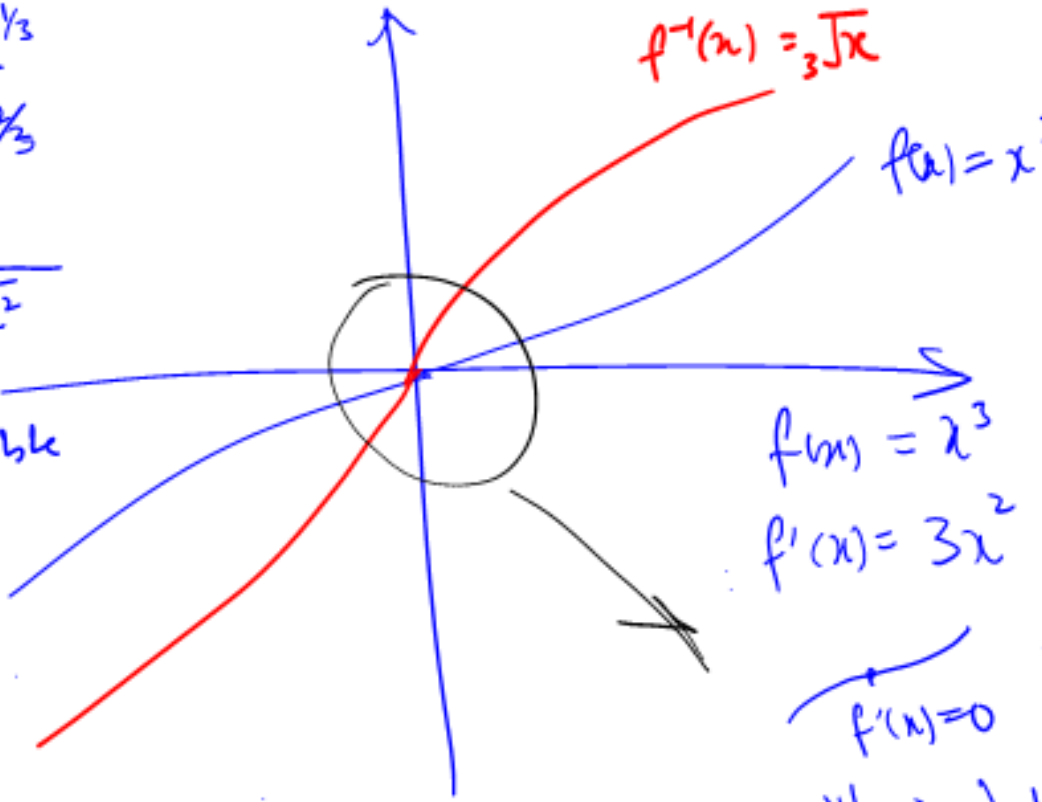
Consider the function $f(x) = x^3$ with the inverse $f^{-1}(x) = \sqrt[3]{x}$.

Let $g(x) = \sqrt[3]{x} = x^{1/3}$

$g'(x) = \frac{1}{3}x^{-2/3}$

$= \frac{1}{3\sqrt[3]{x^2}}$

g is not differentiable at $x = 0$.



Derivatives of Inverses of Differentiable Functions

The derivative rule for inverses

Let f be a function with the domain over the interval I. Suppose that f is differentiable and $f'(x) \neq 0$ for all $x \in I$. Then f^{-1} is differentiable over its domain, $\mathrm{Range(f)}$ and the value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value f' at the point $a = f^{-1}(b)$, i.e,

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

or

$$\frac{df^{-1}}{dx}\Big|_{x=b} = \frac{1}{\frac{df}{dx}\Big|_{x=f^{-1}(b)}}$$

$b \in \text{Range}(f)$ / $\text{Domain}(f^{-1})$

Derivatives of Inverses of Differentiable Functions

To show f^{-1} is differentiable is hard (omitted in SIM 1003)

To show $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$, we use

$$f\left[f^{-1}(x)\right] = x$$

$$\frac{d}{dx}\left[f\left[f^{-1}(x)\right]\right] = \frac{d}{dx}(x)$$

$$f'(f^{-1}(x)) \underbrace{\frac{d}{dx} f^{-1}(x)}_{(f^{-1})'(x)} = 1$$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

← valid because $f'(x) \neq 0$ for all x in the domain

Derivatives of Inverses of Differentiable Functions

Example: $f(x) = x^2$, $x > 0$.

From the previous example, $f^{-1}(x) = \sqrt{x} = g(x)$

$f'(x) = 2x \Rightarrow f'(2) = 2(2) = 4$

$(f^{-1})'(x) = g'(x) = \frac{1}{2\sqrt{x}}$

$f'(\sqrt{x}) = 2\sqrt{x}$

$= \frac{1}{f'(\sqrt{x})}$

At $x = 2$, $f(2) = 4$ ($a \to 2$, $b \to 4$)

$f^{-1}(4) = 2$

$= \frac{1}{f'(f^{-1}(x))}$

$$(f^{-1})'(4) = \frac{1}{f'(f^{-1}(4))}$$

$$= \frac{1}{f'(2)} = \frac{1}{4}.$$

Derivatives of Inverses of Differentiable Functions

Example: Let $f(x) = x^3 - 2$, $x > 0$. Find the value of $\frac{df^{-1}}{dx}$ at $x = 6 = f(2)$ without finding a formula for $f^{-1}(x)$.

$f(x) = x^3 - 2$

$\frac{df}{dx} = 3x^2$

$(f^{-1})^{-1}(x) = f(x)$.

$$\left.\frac{df^{-1}}{dx}\right|_{x=6=f(2)} = \frac{1}{\left.\frac{df}{dx}\right|_{x=2}}$$

$$= \frac{1}{3(2)^2} = \frac{1}{12}.$$